Evalify

ML-DL-Ops (CSL 7120)

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IIT Jodhpur, Semester 2

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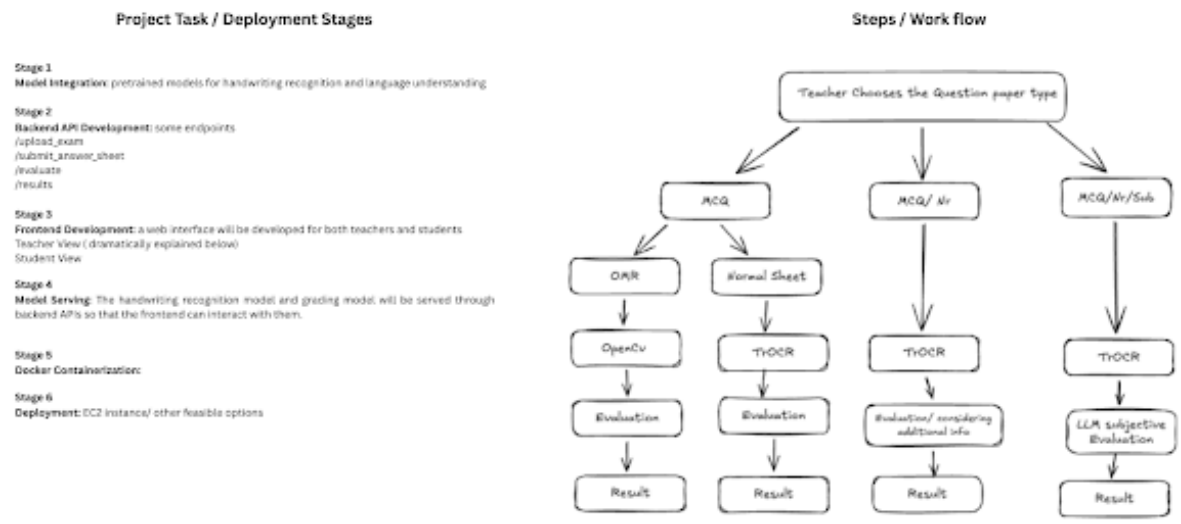
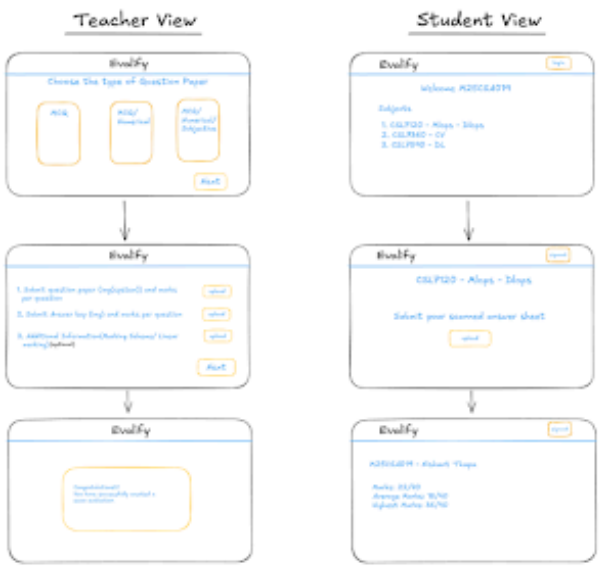
1. Project Poster





Libraries / API

- OpenCV
- StreamLit
- HuggingFace
- Docker
- Python
- Torch
- Javascript
- Html
- AWS
- MongoDB

Dataset / Model

- TrOCR (base-sized model, fine-tuned on IAM)
- Mistral 7B

2. Purpose of project

Evalify is an automated handwritten answer sheet evaluation system that removes the manual grading burden on teachers and TAs. It supports MCQ grading via OMR detection, numerical evaluation with tolerance-based logic, and subjective answer assessment using a locally hosted LLM(Ollama). Built with React, FastAPI, Docker, and MLflow, it aims to deliver fast, consistent, and scalable grading, freeing educators to focus on teaching instead of paperwork.

Feedback Suggested:

Subjective answer evaluation using LLM needs more detail on prompt engineering and grading rubric handling. MongoDB choice could be justified better. Consider adding experiment tracking (MLflow) and model versioning details.

3. Project Phases and component in details

Project is completed in following phases

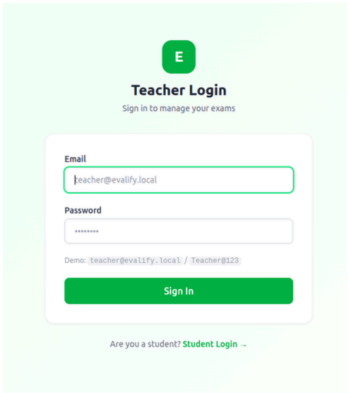
Phase 0 — Project Setup:

Phase 1 — Frontend Shell:

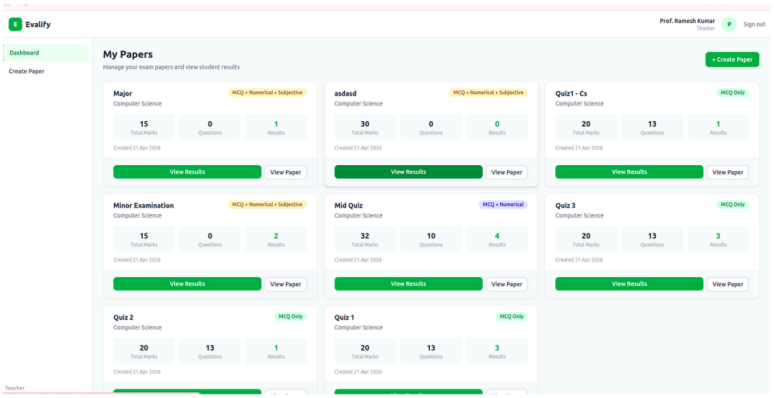
Our project has mainly two view: teacher and student view

Login page

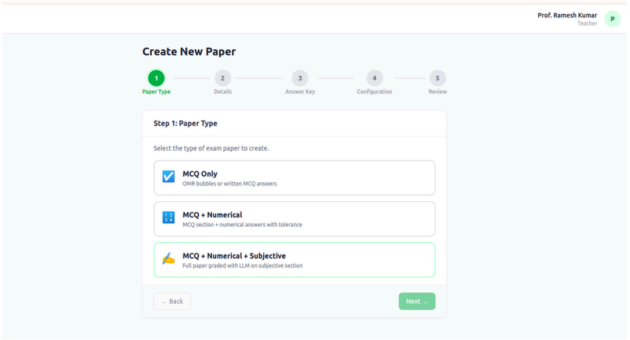
(credentials will be assigned by institute)



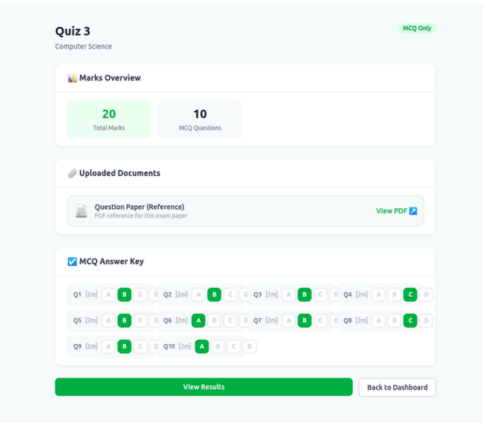
Teacher Dashboard



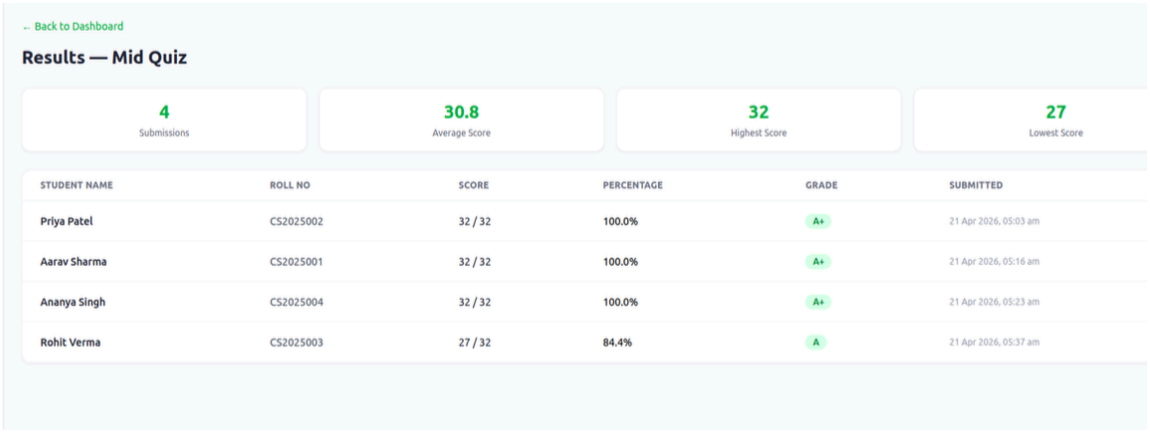
Create a new paper (Three Types Allowed)



Individual Paper View

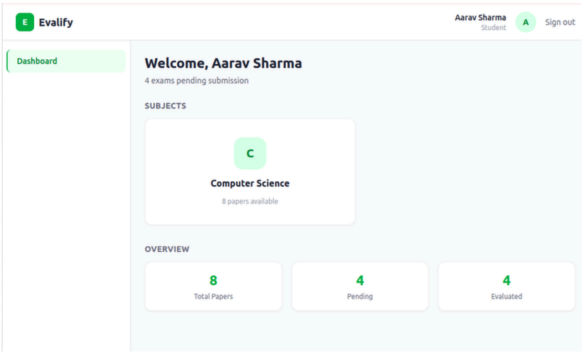


Result view: shows the result of the Students: In future will be consist of several graphs: example normal distribution: which can be used to visualize and analyze performance of class

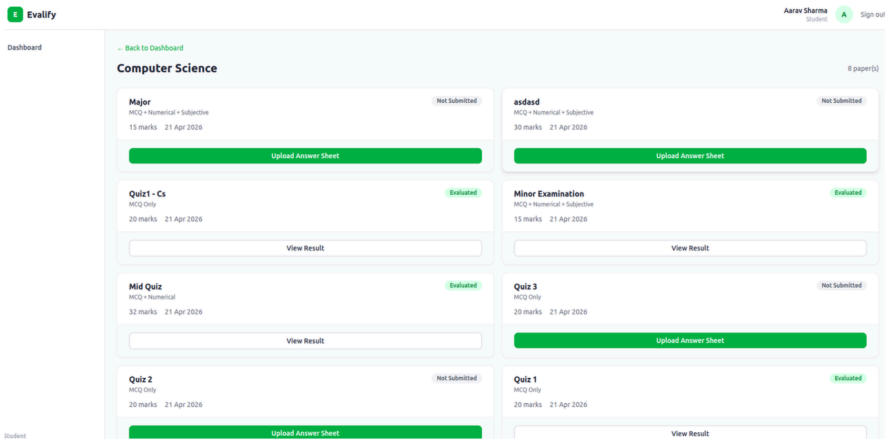


Student View:

Student Dashboard



Subject test

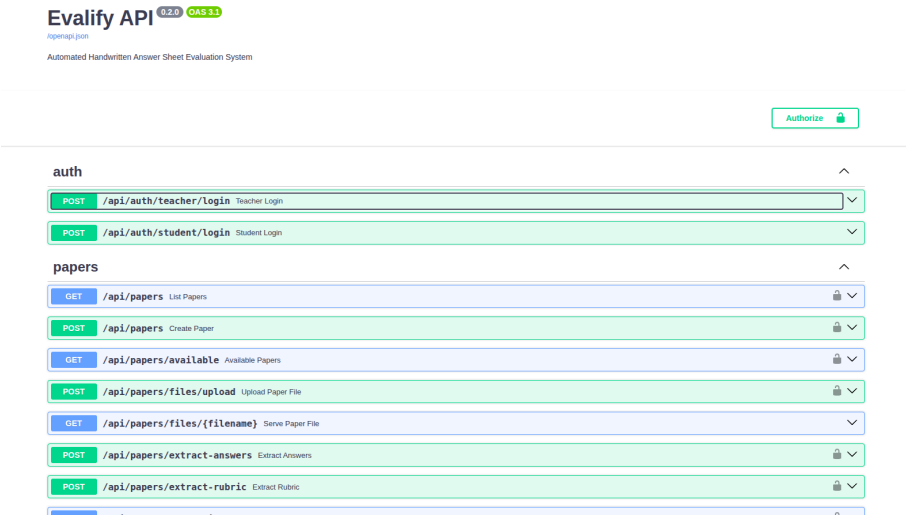


Phase 2 — Backend Foundation

MongoDB: container is used , for authentication, storing the data of the students

Initially we have created 6 dummy students their details are stored in MongoDB database, it is also used for Authentication

SwaggerDocs: is used to Visualize and for detail documentation of all the exposed APIs



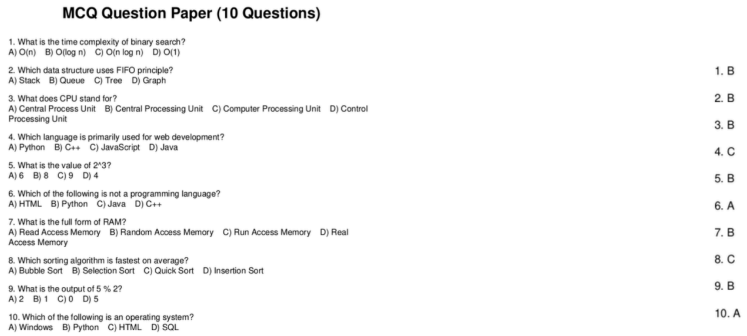
Phase 3 – Type 1: Question Paper implemented (MCQs)

MCQs solution can be given in form of OMR or normal hand written answer sheet

Hence two approach is used to handle both cases

- To read the Students OMR we use Open CV OMR engine is to handle the simple cases, they give very fast response
- For the handwritten answersheet we have use local hosted Ollama 11b model, earlier we decided to use TrOcr model, but it resulted in hallucination of words

Sample MCQs Question Paper and Answer Key



MCQ Question Paper (10 Questions)

1. What is the time complexity of binary search?
A) O(n) B) O(log n) C) O(n log n) D) O(1)

2. Which data structure uses FIFO principle?
A) Stack B) Queue C) Tree D) Graph

3. What does CPU stand for?
A) Central Process Unit B) Central Processing Unit C) Computer Processing Unit D) Control Processing Unit

4. Which language is primarily used for web development?
A) Python B) C++ C) JavaScript D) Java

5. What is the value of 2^3?
A) 6 B) 8 C) 9 D) 4

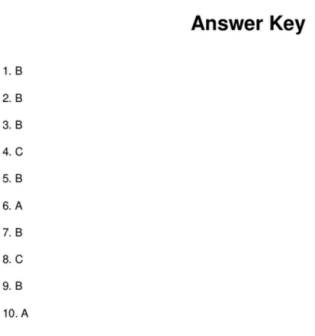
6. Which of the following is not a programming language?
A) HTML B) Python C) Java D) C++

7. What is the full form of RAM?
A) Read Access Memory B) Random Access Memory C) Run Access Memory D) Real Access Memory

8. Which sorting algorithm is fastest on average?
A) Bubble Sort B) Selection Sort C) Quick Sort D) Insertion Sort

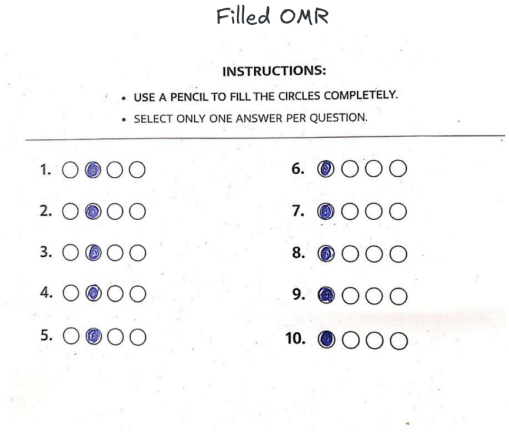
9. What is the output of 5 % 2?
A) 2 B) 1 C) 0 D) 5

10. Which of the following is an operating system?
A) Windows B) Python C) HTML D) SQL



Answer Key

1. B
2. B
3. B
4. C
5. B
6. A
7. B
8. C
9. B
10. A

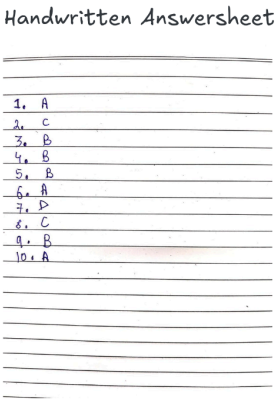


Filled OMR

INSTRUCTIONS:

- USE A PENCIL TO FILL THE CIRCLES COMPLETELY.
- SELECT ONLY ONE ANSWER PER QUESTION.

1. ○ ● ○ ○
2. ○ ● ○ ○
3. ○ ● ○ ○
4. ○ ● ○ ○
5. ○ ● ○ ○
6. ● ○ ○ ○
7. ● ○ ○ ○
8. ● ○ ○ ○
9. ● ○ ○ ○
10. ● ○ ○ ○



Handwritten Answersheet

1. A
2. C
3. B
4. B
5. B
6. A
7. D
8. C
9. B
10. A

Phase 4 - Type 2 MCQ + Numerical Question Paper

- Main crux of this type of question paper was that the Numerical Answers can have more than one solution
- Lets see a template prompt send to the olama model to understand the Student Answer sheet

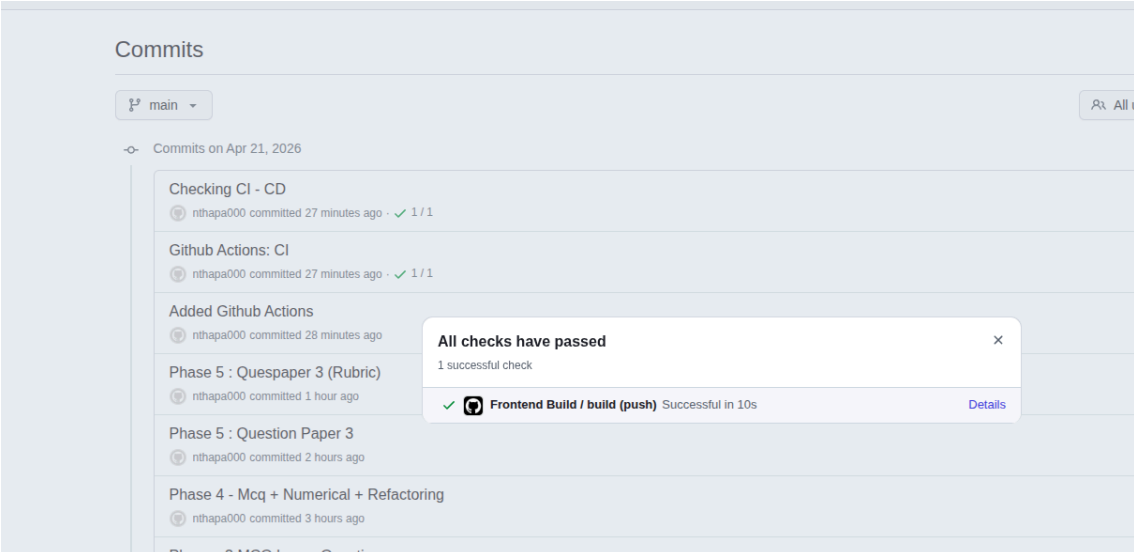
```
) -> str:
    """
    Build the combined extraction prompt for a Type 2 answer sheet.

    Students write a single answer sheet with all questions numbered Q1..Q_total.
    - Q1  …  Q{mcq_count}                          → MCQ section (letter answers)
    - Q{mcq_count+1} … Q{mcq_count+numerical_count} → Numerical section (number/expression answers)

    The prompt uses this natural Q-numbering so Ollama matches what it sees on the sheet.
    """
    total = mcq_count + numerical_count
    num_start = mcq_count + 1

    return f"""You are reading a student's answer sheet with {total} questions total.

Questions Q1 to Q{mcq_count} are MCQ questions — answers are letters ({options}).
Questions Q{num_start} to Q{total} are Numerical questions — answers are numbers or expressions.

Rules for MCQ (Q1–Q{mcq_count}):
- Single option written (e.g. "B")             → value is "B"
- Multiple options written (e.g. "A" and "C")  → join sorted with comma: "A,C"
- Blank or unreadable                           → omit that key

Rules for Numerical (Q{num_start}–Q{total}):
- Copy the number or expression exactly as written (e.g. "3", "3.0", "1/2", "0.5")
- If two values are written separated by comma (e.g. "5/8, 0.625") take the FIRST one
- Blank or unreadable                           → omit that key

Return ONLY a JSON object — no explanation, no preamble.

{{
  "mcq":       {{"Q1": "C", "Q2": "B", ..., "Q{mcq_count}": "B"}},
  "numerical": {{"Q{num_start}": "3.5", "Q{num_start+1}": "1.5", ..., "Q{total}": "3"}}
}}

Now extract all answers from the image."""
```

Phase 5 - Type 3 MCQ, Numerical and Subjective Question Paper:

- Here we implemented a LLM subjective evaluation by doing Prompt engineering and sending high quality prompts as request to locally hosted LLM
- We mainly had two main part of the prompt send for evaluation which was
- Grade Rubric and Reference Answer(Answer referred by LLM for Evaluation)

Phase 6 - Github Actions and Docker containerization

- Implemented simple but very essential checks like npm run build and boosted continuous integration of the code (CI-CD)



- Containerized every component

4. Limitation

- Locally runned LLMs like ollama:11b can only ensure the accuracy upto a certain extent, larger the model used better the accuracy
- Worse the handwriting of the student harder it becomes for LLM to understand it, and accuracy decreases by a lot

5. Github and Demo Link

Github: https://github.com/nthapa000/Evalify

Drive Link: Evalify.mp4